BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Autonomous Dengue Breeding Spot Detector Robot

Group No- 1

Section- 4

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Abstract

The main aim of our project is to build an autonomous robot whose primary function is to find sluggish water. Sluggish water is the perfect place for mosquitoes to breed. We used the Arduino uno as the main controller of the robot. The robot finds stagnant water with the help of a moisture and humidity sensor. It also uses an ultrasonic sensor to move around automatically. After detecting sluggish water, it flashes a LED and sounds a buzzer. It can also save the count of incidents where it finds sluggish water. The main objective of this project is to help our community in finding mosquito breeding spots in the society.

Keywords

Robotics, autonomous, microcontroller, sensor, Arduino

1. Introduction

Dengue is one of the alarming problems in our country. Many people die because of dengue in our country. Dengue mosquitoes breed in stagnant water. The main goal of this project is to build a robot that will identify the spots with stagnant water and sound a buzzer. Arduino processes the data input from the sensors and controls the motor. The robot uses a voting logic system to make sure the data we find is accurate.

2. Related Work/ Inspiration

Many existing systems use large drones to map water. Other methods rely on manual human inspection. Our project is different because it is a small ground

robot. It uses a moisture sensor and a humidity sensor together. Both sensors must agree before an alert happens. This voting logic solves the problem of false alarms.

3. Technical Approach

System Architecture:Arduino Uno is the brain of our robot. It receives physical data from the connected sensors and makes decisions based on our programmed code.

Components Used:

Arduino Uno: The microcontroller, L298N Motor Driver, DC Motors, HC SR04 Sonar, DHT11 Humidity Sensor: A tool to measure water vapor in the air, Moisture Sensor Servo Motor, Buzzer and White LED, Battery: Li-ion Battery 3.7V.

Cost Breakdown:

Arduino Uno: 1000 BDT

Motor Chassis Kit: 800 BDT

Sensors and L298N: 600 BDT

Batteries and Wires: 600 BDT

Beadboard and Motors: 400 BDT

Motors: Driver uses pins 5, 8, 10, and A1.

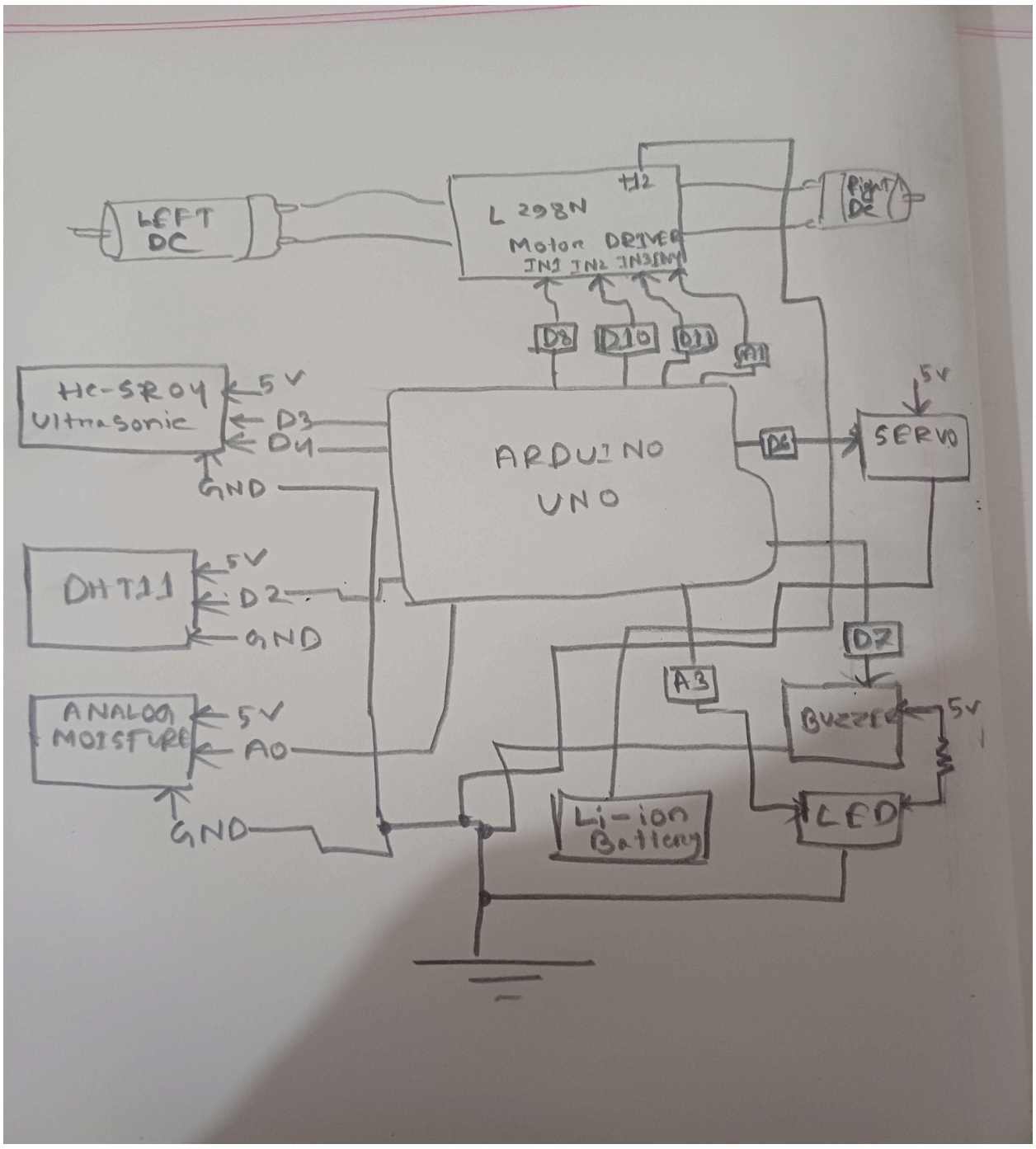
Navigation: Sonar uses pins 3 and 4. Servo uses pin 6.

Sensors: Humidity is on pin 2. Moisture is on pin A0.

Alerts: Buzzer is on pin 7. White LED is on pin A3.

Power: Batteries share a ground wire for stability.

Circuit:


LEFT DC
L 298N
Motor DRIVER
IN1 IN2 IN3 IN4
Right DC
D8
D10
D11
A1
HC-SR04
Ultrasonic
5V
D3
D4
GND
ARDUINO
UNO
D6
SERVO
5V
DHT11
5V
D2
GND
D7
A3
ANALOG
MOISTURE
5V
A0
GND
Buzzer
5V
Li-ion
Battery
LED

Functionality: The robots function is to drive forward automatically. It stops and turns if a wall is closer than 20 centimeters. It checks the air and ground for water. If both sensors detect water the robot stops. The servo lowers the sensors to the ground for three seconds to confirm and if water is really there the LED flashes five times and the buzzer sounds. After that the robot saves this to its memory.

Challenges: Our biggest challenge was power management. The robots wheels were not working at all at first. The 9V battery could not give enough current to the motors. The voltage drop caused the Arduino brain to restart over and over. We solved this by adding a second battery. One battery powers only the motors. The other battery powers only the Arduino. We connected their ground pins together to fix the communication. But it also did not work. The wheels would spin for a while then it would not work. We solved that by using a Li-ion battery. We powered the entire thing with two Li-ion batteries.

4. Sustainability & Impact

Our project inspires a chemical free approach to pest control because it can identify breeding sites before the birth of the mosquitoes. This reduces insecticide spraying. which reduces the harm to local biodiversity. The system uses very low power components like the Arduino Uno and other low power sensors.. The dengue crisis in Bangladesh is immense. It can inform residents and building caretakers by providing an automated alert system .

Our project can be implemented in a different manner as an example, a UAV version which can make the task easier and more convenient.That can detect the breeding spots as well as work as a pest control unit .

The moisture sensor must be very close to the ground, which may cause physical damage.

5. Results & Discussion:

The robot was tested in a controlled environment. We ran it to out BRACU rooftop. The ultrasonic sensor successfully maintained a 20cm buffer from obstacles. It identifies the moisture. The bright sunlight makes the LEDs a bit hard to see, that is why the buzzer is there to help..

6. Conclusion:

We built an autonomous robot to detect stagnant water, so that we can detect dengue breeding spots. It uses sensor fusion and avoids obstacles perfectly. This project shows how robotics can solve real health problems.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Iftequr Ahmed	22299172	Hardware design, Components selection, Building the robot.	Project idea selection and implementation, managing the project, distributing work.
Md. Nafizul Sadik	21201313	Helping in Circuit design and hardware implementation.	Connecting sensors, testing sensors.

Shafi Ahmed Adib	23101307	Designing the circuit.	Building the chassis of the Robot, wiring the components.
Sadat Hossain	22201314	Code creation	Alternate code creation after bug found
Fatin Rahman Taseen	22299002	Code Testing, Design planning.	Bug finding

References

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